

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

MODULE - 45

CLIMATE CHANGE - 6

MEASURES FOR MANAGEMENT OF SOIL ORGANIC CARBON (SOC)

3 RESIDUE MANAGEMENT & MULCHING

CONSEQUENCES OF ON-FARM BURNING OF CROP RESIDUES

- Burning of crop residues leads to release of soot particles and smoke causing health problems.
- It also leads to emission of greenhouse gases (carbon dioxide, methane and nitrous oxide) causing global warming and loss of plant nutrients like N, P, K and S.
- Heat generated from the burning of crop residues elevates soil temperature causing death of beneficial micro-organisms.
- Repeated burnings in a field diminish the microbial population permanently.



CLIMATE CHANGE

MULCHING

- Mulching is the process of covering bare soil with a porous material to improve the condition of the soil underneath.
- Organic mulches such as agriculture residue and manure will decompose into humus improving the soils' fertility, texture and structure. It will encourage earth worms that aerate the soil, protecting them from soil contaminants and regulating the ground temperature.
- Mulches will help the soil keep its moisture longer by slowing the evaporation process.
- Mulch can reduce the impact of heavy rains and reduce soil erosion and nutrient leaching.
- Inorganic mulch is more permanent than organic mulch but has little benefit to the soil's fertility or texture.
- Inorganic mulches, such as plastic sheets, stones, and gravel chips are longer lasting and odorless but they won't add nutrients to the soil and they may also obstruct water and nutrients from reaching the roots and can suffocate plants.



MEASURES FOR MANAGEMENT OF SOIL ORGANIC CARBON (SOC)



Learning

4

CROP ROTATION AND CROP DIVERSIFICATION

Learning Space

- Crop rotation is the practice of growing different crops in succession on the same land to preserve the productive capacity of the soil.
- Crop diversification means growing a variety of crops in an area, not just one. If one crop fails this year, the area can still survive.

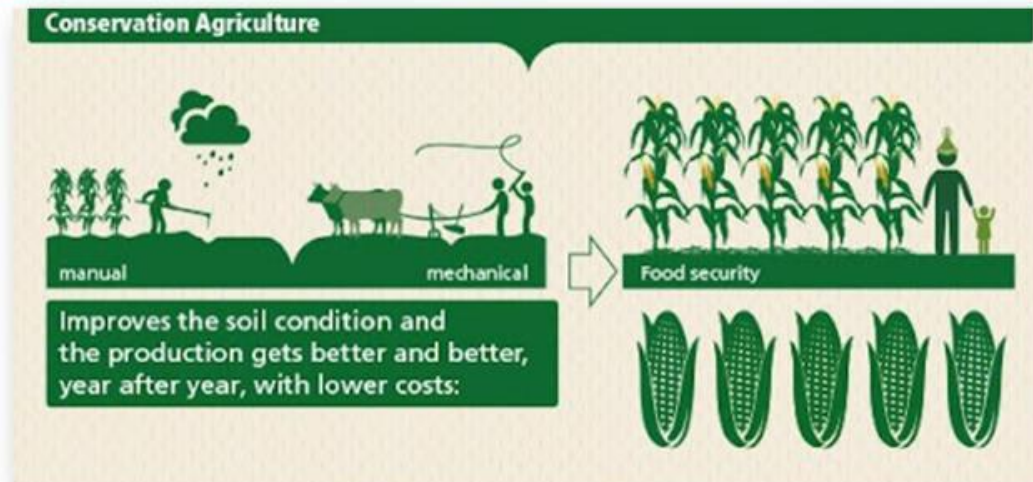


Learning Space

SUSTAINABLE AGRICULTURE PRACTICES

CONSERVATION AGRICULTURE (CA)

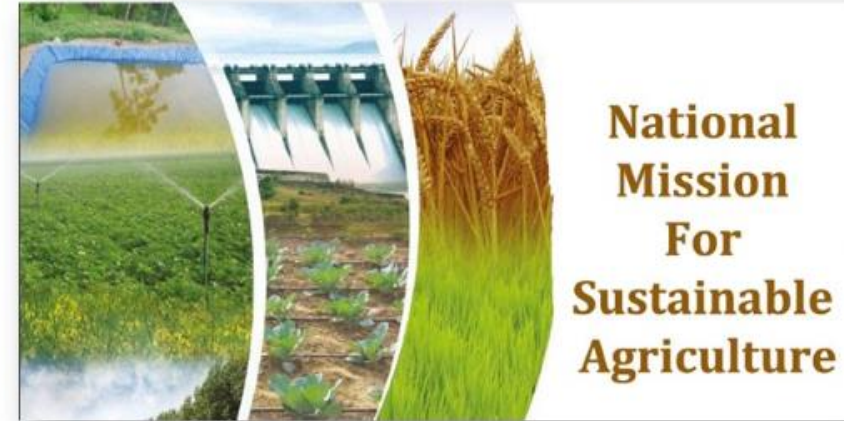
- Soil disturbance regulation, surface residue management and crop rotation are the fundamentals (core pillars) of CA.
- Minimum physical soil disturbance on cropland– zero or reduced tillage enhances soil natural processes and preserves the soil structure.
- Crop residue soil cover controls erosion and temperature effect on surface soil, provides substrate for microorganism existence.
- Soils under diverse cropping systems by and large have a higher SOC content than monocultures.



I NATIONAL MISSION FOR SUSTAINABLE AGRICULTURE

MISSION OBJECTIVES

- To make agriculture more productive, sustainable, remunerative and climate resilient by promoting location specific Integrated Farming Systems;
- To conserve natural resources through appropriate soil and moisture conservation measures;
- To adopt comprehensive soil health management practices based on soil fertility maps, soil testing, judicious use of fertilizers etc.
- To optimize utilization of water resources through efficient water management for achieving 'more crop per drop'.
- To improve productivity of rainfed farming.





I NATIONAL MISSION FOR SUSTAINABLE AGRICULTURE

WHAT IS INTEGRATED FARMING?

- Integrated Farming means to integrate crop production with livestock management which complement each other.
- It is economically viable, profitable and environmentally suitable.
- It involves observation and continuous improvement in all areas of a farming through efficient management practices.
- Integrated Farming combines the best of modern tools and technologies with traditional practices.



CLIMATE CHANGE

I NATIONAL MISSION FOR SUSTAINABLE AGRICULTURE

MISSION STRATEGY

- Promoting integrated farming system covering crops, livestock & fisheries for enhancing livelihood opportunities, ensuring food security and minimizing risks from crop failure.
- Popularizing resource conservation technologies and introducing practices that will support mitigation efforts in times of extreme climatic events or disasters like prolonged dry spells, floods etc.
- Promoting effective management of available water resources and enhancing water use efficiency.
- Creating database on soil resources to facilitate adoption of location and soil - specific crop and soil management practices.
- Involving academic institutions and professionals in developing climate change adaptation and mitigation strategies.



1 NATIONAL MISSION FOR SUSTAINABLE AGRICULTURE

MISSION INTERVENTIONS

- NMSA has following four Programme components or activities:
 1. Rainfed area development (RAD)
 2. On farm water management (OFWM)
 3. Soil health management (SHM)
 4. Climate change and sustainable agriculture: monitoring, modeling and networking (CCSAMMN)



1 NATIONAL MISSION FOR SUSTAINABLE AGRICULTURE



MISSION INTERVENTIONS



1 RAINFED AREA DEVELOPMENT (RAD)

- This component introduces appropriate farming systems by integrating multiple components of agriculture such as crops, horticulture, livestock, fishery and forestry.
- And soil test/soil health card based nutrient management practices, farmland development, resource conservation and crop selection suitable to local agro-climatic condition are also promoted under this component.

2 ON FARM WATER MANAGEMENT (OFWM)

- OFWM focuses primarily on enhancing water use efficiency by promoting efficient on farm water management technologies and equipment.
- It will also focus on effective harvesting & management of rainwater. Assistance will be extended for adopting water conservation technologies.



1 NATIONAL MISSION FOR SUSTAINABLE AGRICULTURE



MISSION INTERVENTIONS



3 SOIL HEALTH MANAGEMENT (SHM)

- SHM aims at promoting location and crop specific sustainable soil health management which includes: residue management, organic farming practices, encouraging judicious application of fertilizers and minimizing the soil erosion/degradation.
- This component will also provide support to reclamation of problem soils (acidic and saline).

4 CLIMATE CHANGE AND SUSTAINABLE AGRICULTURE: MONITORING, MODELING AND NETWORKING (CCSAMMN)

- Through CCSAMMN research on climate change adaptation/mitigation measures to develop climate smart and sustainable agricultural practices is conducted. and integrated farming system suitable to local agro - climatic conditions.

**Sustainable
Agriculture
Program**

Fighting rural poverty.



1 NATIONAL MISSION FOR SUSTAINABLE AGRICULTURE



SUMMARY



- **National Mission on Sustainable Agriculture**
 1. Diversification of cropping systems.
 2. Promotion of carbon sequestration in agricultural practices and building resilience in soil.
 3. Sustainable soil management practices.
 4. Water saving technologies and
 5. Climate responsive research programmes.



WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

98499 42299